

RAJA ABRAHEEM RASHID

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LinkedIn

Career Objective

Prospective PhD researcher in Artificial Intelligence and Machine Learning with interests in scalable learning systems, distributed intelligence, and trustworthy AI. My work focuses on bridging theoretical machine learning with practical applications in LLMs, computer vision, and federated learning.

Research Interests

My research interests are broadly in machine learning and artificial intelligence. I am interested in building intelligent systems that can learn effectively, adapt to complex environments, and remain reliable in practice. More specifically, I am interested in areas such as machine learning theory, deep learning, reinforcement learning, distributed and privacy-preserving learning, large language models, trustworthy AI, and efficient optimization methods. Overall, I am motivated by the goal of developing AI methods that are both scientifically grounded and useful for real-world applications.

Education

Brunel University London, London, United Kingdom 2024 – 2026
MSc in Artificial Intelligence

- Dissertation: Agentic AI for Autonomous Financial Analysis: Design, Implementation, and Evaluation of a Fully Functional LLM-Powered Equity Research System
- Relevant Coursework: Advanced Deep Learning, Statistical Learning Theory, Convex Optimization, Information Theory

COMSATS University Islamabad, Islamabad, Pakistan 2020 – 2024
BEng in Electrical Engineering (AI Specialization)

- Final Year Project funded by NESCOM: Intelligent Reconfigurable Surfaces for UAV Communications — sole undergraduate recipient of NESCOM Research Funding

Employment History

Curium, Singapore (Remote) Feb 2026 – Present
AI/ML Engineer

- Curium pioneers Continuous Dynamic Calibration™ (CDC) — a proprietary multi-sensor calibration platform spanning LiDAR, Radar, and Camera modalities — enabling rapid escalation of autonomous systems to Level 4 and 5 automation.
- Developing ML pipelines and AI algorithms for automated multi-sensor calibration, ensuring accuracy and reliability of perception data across ADAS and autonomous vehicle deployments.
- Integrating AI/ML-driven calibration intelligence within Curium's CDC platform for robust performance under dynamic environmental and operational conditions.

Telecommunications Research Lab (TRL), IBA, Pakistan (Remote) Jul 2025 – Present
AI Researcher & Co-Investigator (Promoted)

- Promoted from Research Associate to AI Researcher and Co-Investigator within months of joining, in recognition of independent research capability and consistent contribution to the lab's publication pipeline. The group targets exclusively A* conference venues and top-tier IEEE/ACM journals.
- Designing multi-frequency hierarchical learning frameworks that jointly address scalability, client heterogeneity, and catastrophic forgetting in federated settings.
- Developing a nested federated unlearning scheme enabling selective knowledge removal without full model retraining, advancing compliance with data-rights legislation.
- Extending Nested Policy Learning to child-AI interaction, where developmental stage gates safety constraints at multiple policy timescales.
- Collaborating with researchers ranked in the top 2% globally based on citation impact; co-authors span the United Kingdom, United States, UAE, Pakistan, Saudi Arabia, and Oman.

National Cyber Emergency Response Team (NCERT), Government of Pakistan *Feb 2025 – Jun 2025*
AI Research Engineer

- Contributed to the design and development of Pakistan's first AI-powered Security Operations Centre (SOC) toolset — a strategically critical initiative for deployment across all governmental organizations, designed to autonomously process and triage 25 million+ logs per day.
- Designed an end-to-end AI-powered SIEM pipeline: from large-scale log ingestion through LLM-based threat classification to automated alert routing, replacing manual SOC Level 1 analyst operations at national scale.
- Recognized as the youngest engineer selected for Pakistan's National AI Research Team; received a formal Letter of Appreciation from the Director General, NCERT.

Neura-X AI Consulting, London, UK
Technical Lead, AI Systems

Nov 2024 – Dec 2025

- Directed the development of production-grade AI systems spanning computer vision, speech intelligence, and enterprise Retrieval-Augmented Generation (RAG).
- Conducted independent research on efficient transformer architectures for long-context and irregular time-series data in cybersecurity contexts.

IQ Center Ltd, London, UK
Head of AI Education Department

Nov 2024 – Present

- Leading AI curriculum development and public seminars for youth AI education.

Research Publications

Submitted/Under Review

- Raja Abraheem Rashid Ejaz, Faisal Iradat, Waseem Iqbal, and Awais Ahmad. "Nested Learning with Attention-Guided Multi-Frequency Supervision for Hepatic Vessel Segmentation." (*Neural Computing and Applications*, Q1, Under Review).
- Raja Abraheem Rashid Ejaz, Nour Aburaed, and Faisal Iradat. "Nested Learning for Multi-Timescale Electric Vehicle Charging Coordination: A Physics-Informed Deep Reinforcement Learning Framework with Hard Voltage Guarantees." (*IEEE Transactions on Smart Grid*, Under Review).
- Raja Abraheem Rashid Ejaz, Yawar Abbas Bangash, Faisal Iradat, and Muhammad Kumail. "SafeNest: Nested Policy Learning for Multi-Timescale Safety in Child-AI Interaction." (*Scientific Reports*, Under Review).

- Raja Abraheem Rashid Ejaz, Faisal Iradat, and Muhammad Kumail. “A Multi-Timescale Safety Architecture for Developmentally Adaptive Child-AI Interaction via Nested Policy Learning.” (*UKCI 2026, Under Review*).
- Raja Abraheem Rashid Ejaz, Muhammad Waseem Iqbal, Faisal Iradat, and Muhammad Kumail. “Nested Learning for Adaptive-Fidelity ECG Classification: Matryoshka Representation Learning Across the Wearable-to-Cloud Continuum.” (*IEEE Xplore / IEEE Journal, Under Review*).
- Raja Abraheem Rashid Ejaz, Mariam Deda, and Faisal Iradat. “Pre-Encryption Multi-layered Ransomware Detection Framework.” (*ARES 2026 – 21st International Conference on Availability, Reliability, and Security, Under Review*).
- Raja Abraheem Rashid Ejaz, Qazi Muhammad Waiz, and Faisal Iradat. “HADES: Hierarchical Autonomous Digital Evidence System Using Skill-Driven Multi-Agent Orchestration for Digital Forensics.” (*WI 2026, Linz, 6–10 September 2026, Under Review*).
- Raja Abraheem Rashid Ejaz, Faisal Iradat, Waseem Iqbal, and Majdi Mansouri. “Nested Federated Learning: Layer-Wise Multi-Frequency Synchronization for Privacy-Preserving Distributed Intelligence.” (*Cognitive Computation, Under Review*).
- Raja Abraheem Rashid Ejaz, Faisal Iradat, Waseem Iqbal, and Majdi Mansouri. “It’s the Method, Not the Model: Explanation Collapse Under Temporal Malware Drift.” (*IEEE Open Journal of the Communications Society, Under Review*).
- Eaman Tariq, Eaqa Tariq, Izza Sohail Khan, Raja Abraheem Rashid Ejaz, and Faisal Iradat. “From AI to Generative AI: How Emerging Technologies Are Reshaping Public Trust.” (*WI 2026, Under Review*).
- Muhammad Sharique Baig, Raja Abraheem Rashid Ejaz, and Faisal Iradat. “The Impact of Female Under-representation in Cybersecurity on Women’s Vulnerability to Cybercrime.” (*Gender, Technology and Development, Under Review*).

In Preparation

- Raja Abraheem Rashid et al. “Nested Federated Unlearning: A Multi-Frequency Optimization Framework for Privacy-Preserving Distributed Intelligence.” To be submitted to *IEEE Transactions on Artificial Intelligence*.
- Raja Abraheem Rashid and Faisal Iradat. “Employment and Job Retention in the AI Era in LMICs – A Scoping Study.”
- Raja Abraheem Rashid and Faisal Iradat. “Psychological and Ethical Factors Affecting Student Acceptance of AI-Generated Academic Work.”
- Raja Abraheem Rashid et al. “A Nested Learning-Based Framework for Early Environmental Change Detection Using Multi-Temporal Remote Sensing Data.” To be submitted to *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*.
- Raja Abraheem Rashid et al. “Evaluating LLM-Generated Equity Research: A Benchmark Study of Factual Accuracy, Hierarchical Reasoning Quality, and Hallucination Rates in AI-Driven Financial Reporting.” (In preparation; arising from MSc dissertation on agentic financial analysis).

Research Supervision (Co-Supervisor)

Co-supervising undergraduate research projects at Institute of Business Administration (IBA), Karachi in collaboration with Dr. Faisal Iradat. All projects are being prepared for submission to top-tier Q1 journals and A conferences.*

- **FedSRIMU: Spectral-Regularised Federated Unlearning for IoT-Based Intrusion Detection.**
- **Panopticon: A Proposal for Privacy Preservation in Distributed Information Systems.**
- **HADES: Hierarchical Autonomous Digital Evidence System Using Skill-Driven Multi-Agent Orchestration for Digital Forensics.**
- **TEMPEST: Temporal Execution Modeling for Pre-Encryption Ransomware Detection with Timestep-Level Explainability.**

MSc Dissertation

Agentic AI for Autonomous Financial Analysis: Design, Implementation, and Evaluation of a Fully Functional LLM-Powered Equity Research System

Brunel University London, 2025–2026

- Addresses the challenge of automating high-fidelity financial analysis using LLMs within an agentic multi-agent pipeline. The system retrieves live market data, performs fundamental and technical analysis, synthesizes structured equity research reports, and self-evaluates factual consistency.
- A companion research paper benchmarks hallucination rates, hierarchical reasoning quality, and factual accuracy of LLM-generated equity research against analyst-produced ground truth, contributing both an open benchmark dataset and a reproducible evaluation framework to the community.

Selected Technical Projects

Thermal Imaging for Industrial Monitoring

Neura-X, 2024

- Developed a hybrid ViT+CNN architecture with Spatio-Temporal Graph Neural Networks for sand-level monitoring in industrial separators. Achieved 90–95% accuracy via contrastive learning. Key insight: reduced ViT patch size from 16×16 to 4×4 for effective low-resolution thermal adaptation.
- **Stack:** PyTorch, Timm, PyTorch Geometric, OpenCV, Docker.

ZazuAI: On-Premise Meeting Intelligence

Neura-X, 2024

- Built a dual-pipeline speech system combining real-time transcription (Whisper.cpp + DIART, <500ms latency) with high-accuracy post-processing (WhisperX, 96% accuracy, 80% diarization). Automated meeting minutes via Llama.cpp.
- **Stack:** Python, Whisper.cpp, WhisperX, PyAnnote, Llama.cpp, Docker.

Enterprise RAG: Multi-Agent Architecture

Neura-X, 2024–2025

- Built a self-hosted retrieval-augmented generation system using hierarchical multi-agent orchestration and hybrid retrieval (dense + sparse + re-ranking). Improved end-to-end accuracy from 76% (dense-only) to 89% at sub-200ms latency.
- **Stack:** Python, gRPC, Llama.cpp, Sentence-BERT, Qdrant, Elasticsearch, Redis, Docker, Kubernetes.

LLM-Powered SIEM

Government of Pakistan / NCERT, 2025

- First national deployment of an LLM-based SIEM across 200+ federal organizations. Entire pipeline — from data ingestion via Kafka to quantized inference and automated alert routing — designed and implemented by the research team.
- **Stack:** PyTorch, Hugging Face, DeepSpeed, Kafka, Docker, Kubernetes.

Technical & Language Skills

- **Languages:** English (Highly Proficient); Urdu (Native)
- **Programming:** Python (Advanced), C++ (Intermediate), SQL, Bash
- **ML/DL Frameworks:** PyTorch, TensorFlow, JAX, Hugging Face Transformers, scikit-learn, XGBoost
- **LLMs:** Fine-tuning (LoRA, QLoRA), Prompt Engineering, RAG Systems, Quantization (INT8/INT4)
- **MLOps:** Docker, Kubernetes, MLflow, Weights & Biases, Git, CI/CD (GitHub Actions)
- **Data & Infrastructure:** Pandas, NumPy, Kafka, Redis, PostgreSQL, Elasticsearch, Qdrant (Vector DB)
- **Cloud/Distributed:** AWS (EC2, S3), Linux, DeepSpeed, FSDP, Distributed Training
- **Research Focus:** Nested Learning, Federated Learning, Model Unlearning, Agentic AI, Efficient Transformers

Awards and Achievements

- **Selected Participant**, CISPA Summer School 2026, Helmholtz Center for Information Security — presented research in Federated Learning (2026)
- **Youngest Engineer** selected for Pakistan's National AI Research Team, NCERT (2025)
- **Letter of Appreciation** from the Director General, National Cyber Emergency Response Team (2025)
- **NESCOM Research Funding** — sole undergraduate recipient for intelligent UAV communications research (2024)
- **Research Leadership and Supervision** — Co-supervising multiple undergraduate research projects at IBA Karachi, with submissions targeted at Q1 journals and A* conferences

References

Dr. Faisal Iradat — *Research Supervisor & Mentor*

Head, Telecommunications Research Lab (TRL), Institute of Business Administration (IBA), Pakistan.

Dr. Haider Abbas — *DG / Head of National CERT Pakistan*

National Computer Emergency Response Team (CERT), Pakistan.

Dr. M. Muhammad Waseem Iqbal — *Assistant Professor*

Department of Electrical and Computer Engineering, Sultan Qaboos University, Muscat, Oman.